

Specifying Complete Programs

These are the temperature rooms of ITB:

If the temperature is below 21 C, the heater should turn ON, and if the temperature is 21 C or above, the heater should remain OFF.

Write a program specification for this temperature behaviour by providing a **precondition** describing a valid input values for i_1 and a **postcondition** specifying the required output o , which can be either ON or OFF.

$$\{0 \leq i_1 \leq 100\}$$

P

$$\{(i_1 < 21 \rightarrow o = \text{ON}) \wedge (i_1 \geq 21 \rightarrow o = \text{OFF})\}$$

MCQ - 7

7. We are implementing a safety-critical system that uses both hardware and software to achieve its functionality. Since it is designated as safety-critical, it is vital that we are able to guarantee that the system implements its functionality so that it delivers "safe" results as often as possible, even when in some cases we cannot guarantee "correct" behaviour – in the face of hardware malfunction etc.

Let us examine an example. In a function in the software, we need to determine whether to "trip" or not. The behaviour is governed by the tabular expression below.

c_trip : (e_notTrip, e_trip) Note that the safe state here would be $c_trip = e_trip$

m_temp: temperature in degrees C

k_setpoint in range [23°C, 28°C]

k_hys in range [2°C, 5°C]

Initial condition: $c_trip = ?$

	c_trip
$k_setpoint \leq m_temp$	e_trip
$k_setpoint - k_hys < m_temp < k_setpoint$	No Change
$m_temp \leq k_setpoint - k_hys$	e_notTrip

One choice we have to make at the requirements level is what the initial value of c_trip should be. Either e_notTrip or e_trip.

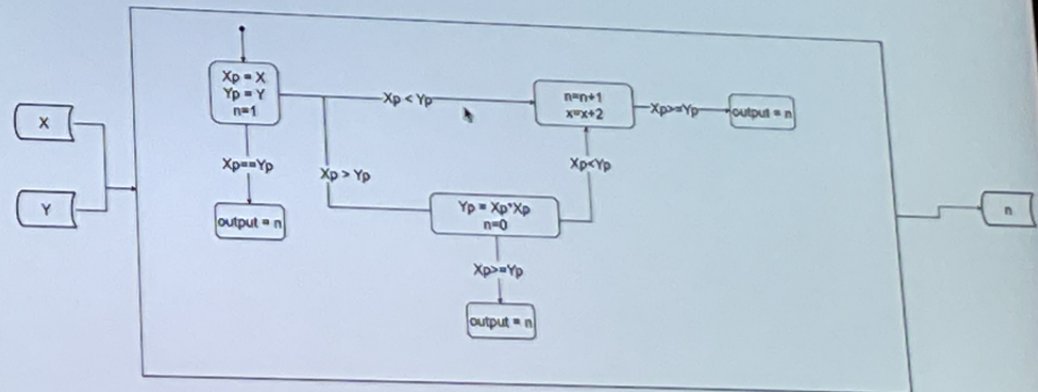
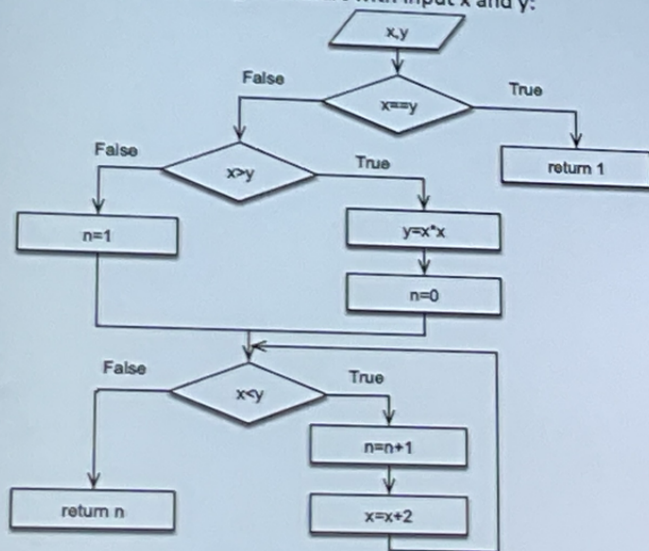
When we implement this behaviour we have further choices to make. For example, we can implement c_trip as a Boolean variable with value True or False (or it could map directly to e_trip or e_notTrip for example). When we do this, we need to consider whether we want to represent Boolean variables by a single bit, or by a full "word", 16 bits for example. Now, bearing in mind that we want both safety and correctness, but that safety is of primary importance, which of the following choices is the best?

- Initial value: $c_trip = e_trip$
Represent c_trip by a single bit. 0 $\rightarrow$ e_notTrip; 1 $\rightarrow$ e_trip
- Initial value: $c_trip = e_trip$
Represent c_trip by 16 bits (or more). All bits 0 $\rightarrow$ e_notTrip; Any bit 1 $\rightarrow$ e_trip
- Initial value: $c_trip = e_notTrip$
Represent c_trip by a single bit. 0 $\rightarrow$ e_notTrip; 1 $\rightarrow$ e_trip
- Initial value: $c_trip = e_notTrip$
Represent c_trip by 16 bits (or more). All bits 0 $\rightarrow$ e_notTrip; Any bit 1 $\rightarrow$ e_trip
- None of the above.

* won't be on midterm 'j

Q-11

11. Consider the following flowchart with input x and y :



Create a stateflow (finite state machine) diagram that implements this flowchart without affecting the original inputs. (10)

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Questions & Attendance (1)



Question

Answered



Question

Which of these scenarios represent HIGH cohesion?

☐ A

OB - single player, kitchen on fire

Submission is closed

Presenting

No slides are being presented

Questions & Attendance (1)

Question

Answered

Question

Which of these scenarios represent LOW coupling?

☐ A



B - multiple intersections & roads, smooth flow

Question is closed

Is Model-Driven Engineering a Software Development Life Cycle Model?

☒ A Yes



☐ B No



↳ model driven = abstract idea
= design way of approaching
a stage in the life cycle

Submission is closed

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Questions & Attendance (1)

Question Answered

Question

A highly cohesive module is one where its components are closely related to one another.

☒ A

True

☐ B

False

Submission is closed



1 New unanswered question

Open



Question

In the 4-variable model, which variables does the software directly work with?

☐ A M and C

☐ B I and O

☐ C M_p and C_p

☒ D (B) and (C)



→ not directly used w/ by software
↳ more like internal variables
defined in software

Submission is closed



Question

Which of these is an extract of a semi-formal requirement specification?



A

"The system shall shut down if it overheats"



B

A flowchart of the system



C

For all t : $\text{temp}(t) > 300 \rightarrow \text{shutdown}(t)$



D

"When the temperature is higher than 300 degrees, then the system will shut down"



↳ semi formal : gherkin : when & then
↳ formal → w/ 't'

Submission is closed

Which of the following is a **time-continuous** monitored variable? Select all that apply.

Select All That Apply



A

The position of a car's accelerator pedal (how far it's pressed)



B

The event "driver pressed the brake pedal"



C

The count of passengers entering a subway train



D

The temperature of reactor coolant in a nuclear plant



not always happening

only when entering
→ discrete count

Submission



1 New unanswered question

Open





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Questions & Attendance (2)



Question
Answered



Question Matching
Answered



Question Matching

Match the term on the left with its correct definition on the right?

Drag and drop options on the right-hand side and submit. For keyboard navigation... [Show more](#) ▾

Regulation

Mandatory rules enforced by governments or authorities (e.g., FAA, CNSC).

Standardization

Agreed-upon methods or practices from industry/consortia (e.g., ISO, IEC, IEEE).

Informal Requirements

Specification written in free natural language, easy to read but often ambiguous.

Semi-formal Requirements

Specification written in a structured but not fully formal way (e.g., Gherkin).

Formal Requirements

Specification written in a mathematically precise language that can be formally verified.

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Questions & Attendance (2)

Question
Answered

Question
Answered

Question

Question

Geometric figure E is described by the following equation:

$$ax^2 + by^2 + c = 0$$

Where a, b, and c, are suitable constants

What specification is this?

☐ A Operational

☒ B Descriptive

Submission is closed

Question 1

Match the Parnas' Rational Design Process (RDP) terms on the left with its correct definition on the right

Drag and drop options on the right-hand side and submit. For keyboard navigation... [Show more](#) ▾

Module Guide (MG)



Describes *how* the module's behaviour is implemented



Module Interface Specification (MIS)



Defines what a module *offers* (its interface)



Module Internal Design (MID)



Describes the decomposition hierarchy of modules



Information hiding



Design principle that separates interface from implementation details



Answered

Submissions closed

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Questions & Attendance (2)

Question
Answered

Question
Answered

Question

Which of the following is **NOT** part of a module's interface?

- ☐ A Function signatures
- ☐ B Expected input and output types
- ☒ C Internal data representations
- ☐ D Preconditions and postconditions

Submission is closed



1 New unanswered question

Open



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Questions & Attendance (2)

Question
Answered

Question 2
Answered

Question 2

Match the Object-Oriented terms on the left with its correct definition on the right

Drag and drop options on the right-hand side and submit. For keyboard navigation... [Show more](#)

Class	A kind of ADT that supports instantiation
Object	A variable or instance of a class
Method	Operation defined inside a class
Inheritance	Mechanism for extending behaviour from another class

Answered

Submissions closed

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Questions & Attendance (2)

Question
Answered

Question
Answered

Question

Question

DFDs primarily model:

- ☐ A Control flow between modules
- ☒ B Data flow and transformations between processes
- ☐ C Object relationships
- ☐ D System architecture in code

Submission is closed



1 New unanswered question

Open



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Questions & Attendance (2)

Question
Answered

Question 3
Answered

Question 3

Match the Finite State Machines (FMSs) term on the left with its correct definition on the right

Drag and drop options on the right-hand side and submit. For keyboard navigation... [Show more](#)

State (Q)	Represents a system condition	✓
Input (I)	Causes transition between states	✓
Transition function	Defines how input changes the current state	✓
State table	Tabular representation of transitions	✓

Answered

Submissions closed